



UNIVERSITI PUTRA MALAYSIA

FAKULTI SAINS KOMPUTER DAN TEKNOLOGI MAKLUMAT

LAB 1– Individual

Week 1

OBJECTIVE:

This lab exercise aims to familiarize the student with the basic operation of the Arduino Uno and some fundamental concepts on embedded systems.

DURATION:

1 week / 3 hours

DUE DATE OF SUBMISSION:

Submit your assignment to your demonstrator before the next lab.

Don't forget to sign the assignment submission form or your marks will not be graded.

PROGRAMME OBJECTIVE:

CPS3-3: Apply appropriate algorithms and techniques to computing solutions.

RUBRIC:

CRITERIA	LEVEL 1 Very Poor	LEVEL 2 Poor	LEVEL 3 Good	LEVEL 4 Very Good	LEVEL 5 Excellent
The ability to perform Ohm's Law calculation.	Unable to perform OS tasks or able to perform Ohm's Law calculation with very limited success (< 40%).	Able to perform Ohm's Law calculation with limited success ($\geq 40\%$, < 60%).	Able to perform Ohm's Law calculation with some success ($\geq 60\%$, < 80%).	Able to perform Ohm's Law calculation with considerable success ($\geq 80\%$, < 100%).	Able to perform Ohm's Law calculation with outstanding success (100%).

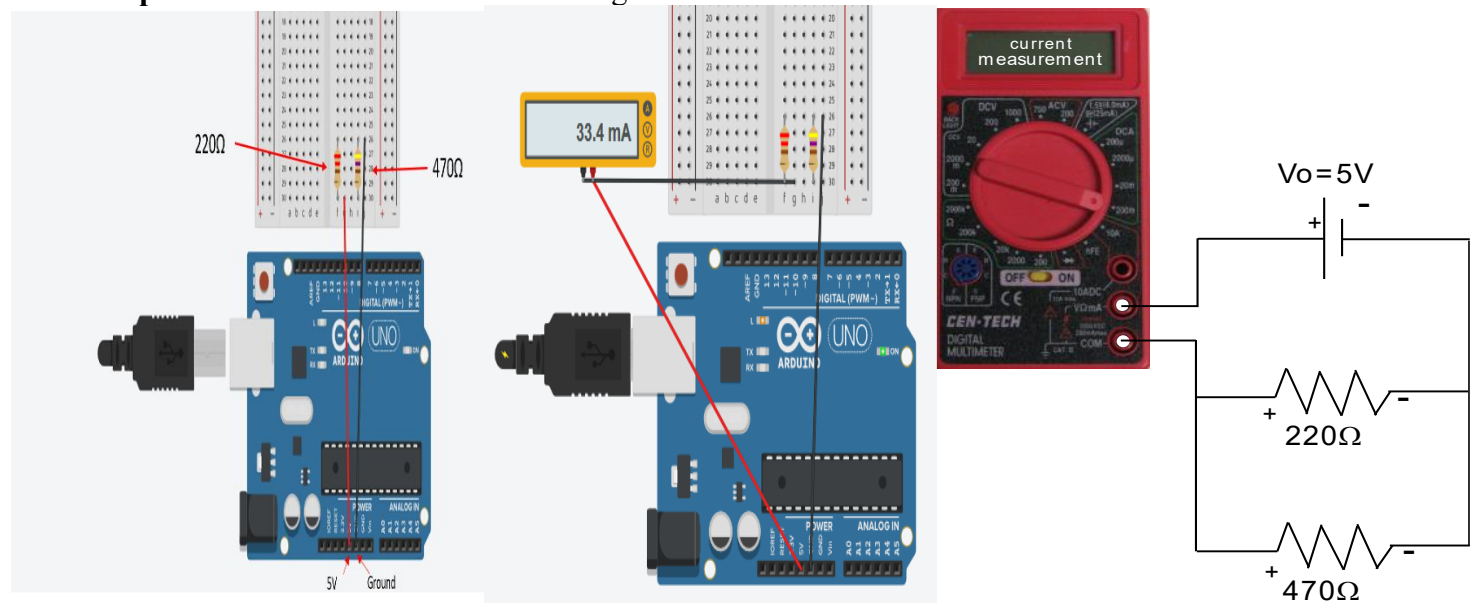
QUESTIONS:

Part A: Ohm's Law, Voltage, Resistance, Current & Introduction to Tinker CAD

1. Visit <https://www.tinkercad.com/>
2. Login and Select TinkerCAD Circuits
3. Familiarize with all the modules and sensors available to you. Play around.

QUESTION

Build the **parallel circuit** as shown below using Arduino Uno board. Use resistors 470Ohm and 220Ohm resistors.



My Own Arduino Board

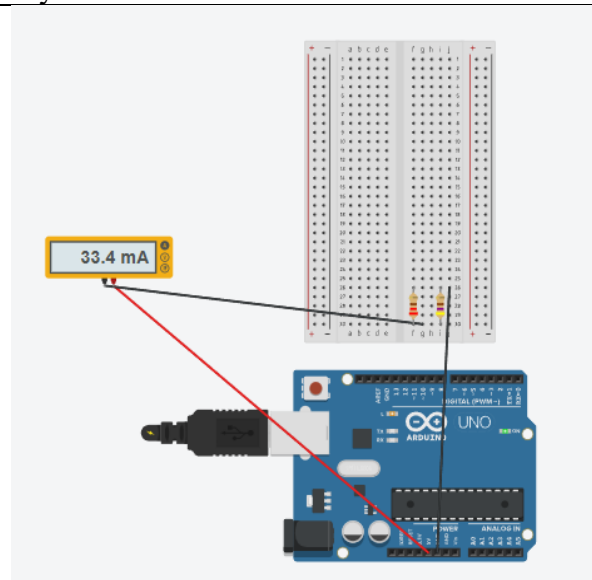


Figure 2: Circuit Diagram

1. Measure and record the total current flowing from the power source in the circuit.
(Be sure to reconfigure your multimeter for measuring current.)



2. Calculate the current after you make your measurement. Build the **parallel circuit**, use resistors 470Ohm and 220Ohm resistors.

- Use Ohm's law:

$$V = I \cdot R \quad \text{or} \quad I = \frac{V}{R}$$

Elements Symbol	RESISTOR	CAPACITOR	INDUCTOR
Denoted by	R	C	L
Equation	$R = \frac{V}{I}$	$C = \frac{Q}{V}$	$L = \frac{V_L}{(di/dt)}$
Series	$R_T = R_1 + R_2$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$	$L_T = L_1 + L_2$
Parallel	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	$C_T = C_1 + C_2$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2}$

$$\frac{1}{R_p} = \frac{1}{470} + \frac{1}{220}$$

$$= 0.00213 + 0.00455 = 0.00668$$

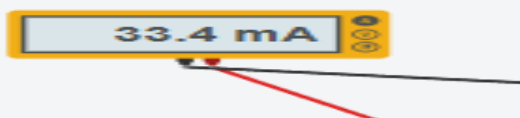
$$R_p = \frac{1}{0.00668}$$

$$= 149.7 \text{ Ohm}$$

$$I_{total} = \frac{5V}{149.7\Omega}$$

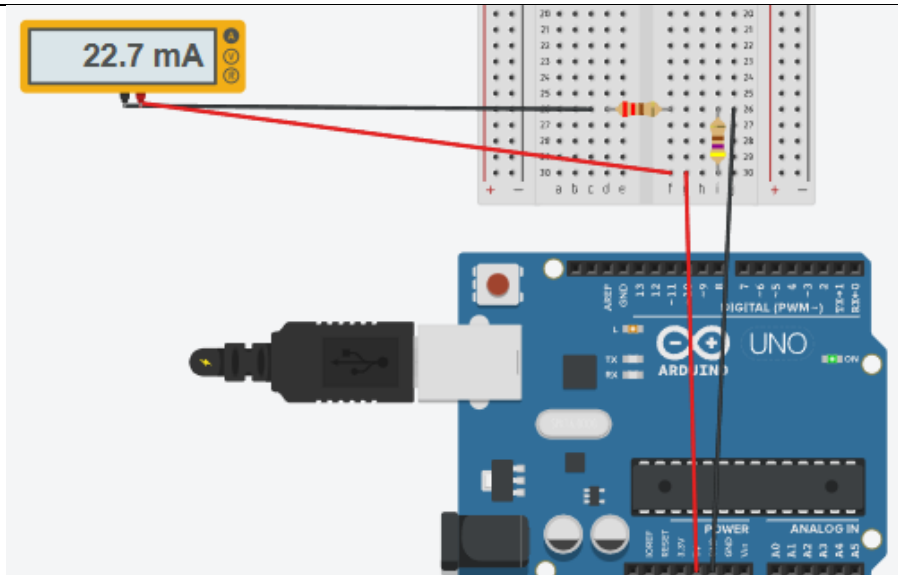
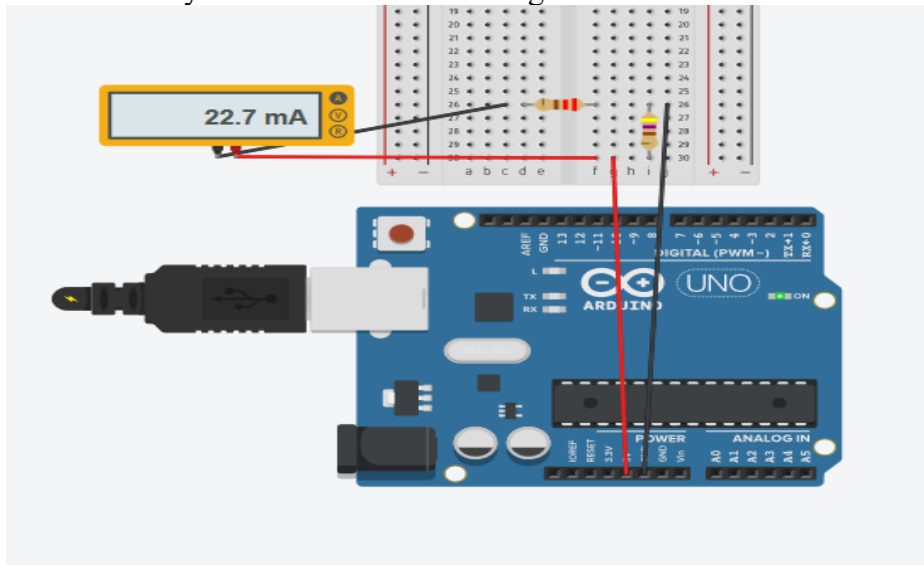
$$= 0.0334A = 33.4 \text{ mA}$$

3. Compare the current values measured and calculated

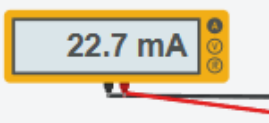


$$= 0.0334A = 33.4 \text{ mA}$$

4. Modify the circuit as shown in Figure 2 below:



5. Measure and record the current through the 220Ω resistor
(Be sure to reconfigure the leads of your multimeter for measuring current.)



6. Calculate the current after you make your measurement

$$I_{220} = \frac{5V}{220\Omega}$$

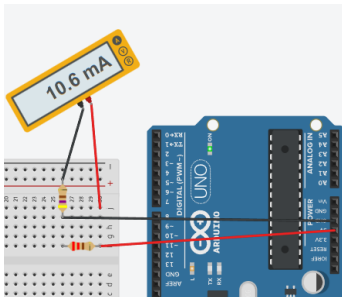
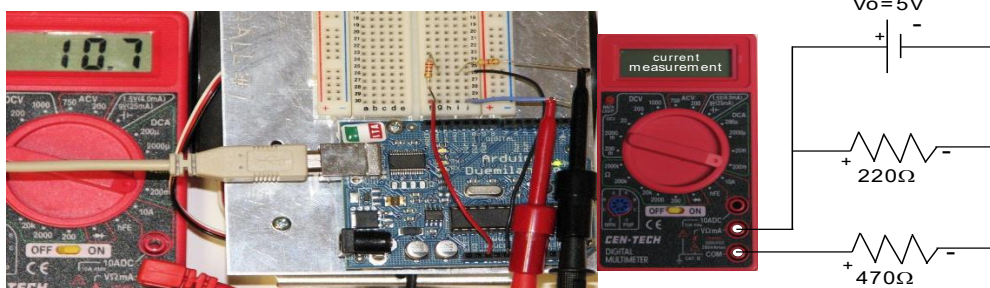
$$I_{220} = 0.0227 \text{ A} = 22.7 \text{ mA}$$

7. Compare the values obtained via measurement and calculation



$$I_{220} = 0.0227 \text{ A} = 22.7 \text{ mA}$$

8. Modify the circuit as below and measure the current through the 470Ω resistor



9. Calculate the current after you make the measurement

$$I_{470} = \frac{5V}{470\Omega}$$

$$= 10.6 \text{ mA}$$

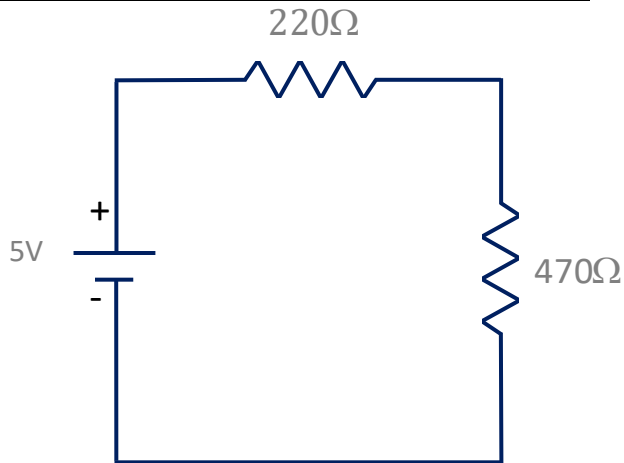
10. Are your findings consistent with Kirchhoff's Current Law? Explain and show calculations.

First part = 33.3 mA

Second Part = 22.7 mA

Third Part = 10.6 mA

$22.7 + 10.6 = 33.3 \text{ mA}$, this matches the first part, therefore Kirchhoff Current Law is correct

Part B (Voltage Drops Around Closed Loops)

$$R_1 = 220\Omega$$

$$R_2 = 470\Omega$$

$$V_1 = 5V$$

1. Find the equivalent resistance

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 220 + 470 = 690 \text{ Ohm}$$

2. Find the current

$$I = \frac{V_{source}}{R_{eq}}$$

$$\frac{5V}{690 \text{ Ohm}} = 0.00725 \text{ A} = 7.25 \text{ mA}$$

3. Find the voltage drop across R_1

$$V_1 = I \times R_1$$

$$V_{R_1} = 0.00725 \times 220 = 1.59 \text{ V}$$

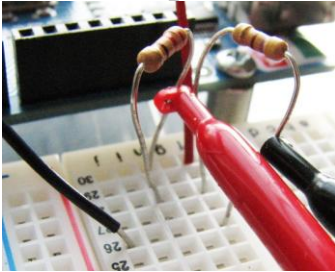
4. Find the voltage drop across R_2

$$V_{R_2} = V_2 = I \times R_2$$

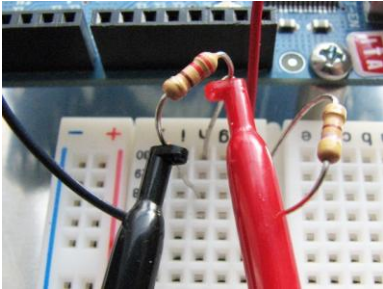
$$0.00725 \times 470 = 3.41 \text{ V}$$

5. Use multimeter to measure voltages around loop

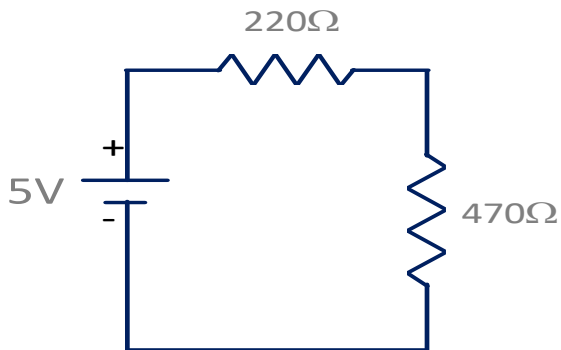
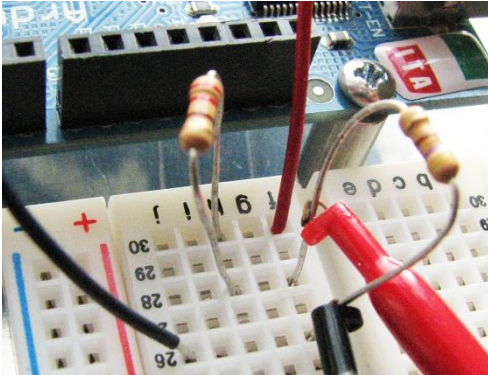
1) From 5V pin to GND $\Delta V1 = 5 \text{ V}$



2) Across the 220Ω resistor $\Delta V2 = -1.59 \text{ V}$



3) Across the 470Ω resistor $\Delta V3 = -3.41 \text{ V}$



6. Compare those measurement with the values you have calculated

The result are the same

7. What is your conclusion regarding the obtained results? Verify them accordingly.

Using Kirchhoff's Voltage Law, sum of voltage on each resistor add up to total voltage

$$1.59 + 3.41 = 5 \text{ V}$$

Part C: First Program Using Arduino (LED Blink)

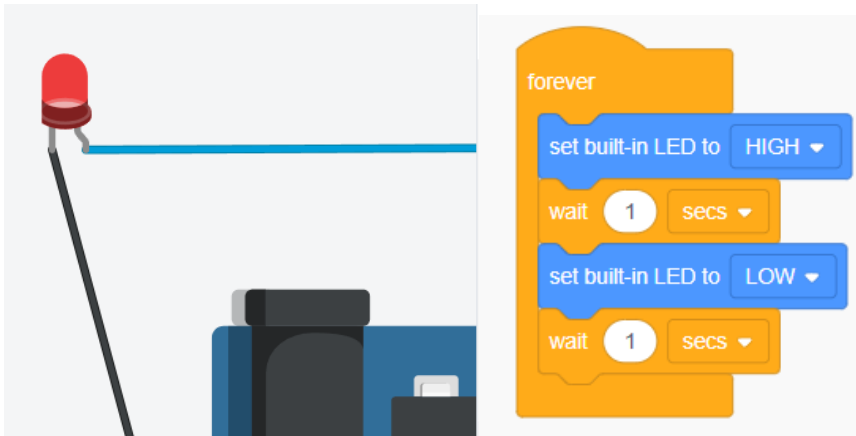
Instruction

1. Connect your Arduino and compile the program sketch below. Explain your observation.

Without any codes, it simply light up the LED. With the code, It now lights up every second and turn off every second.

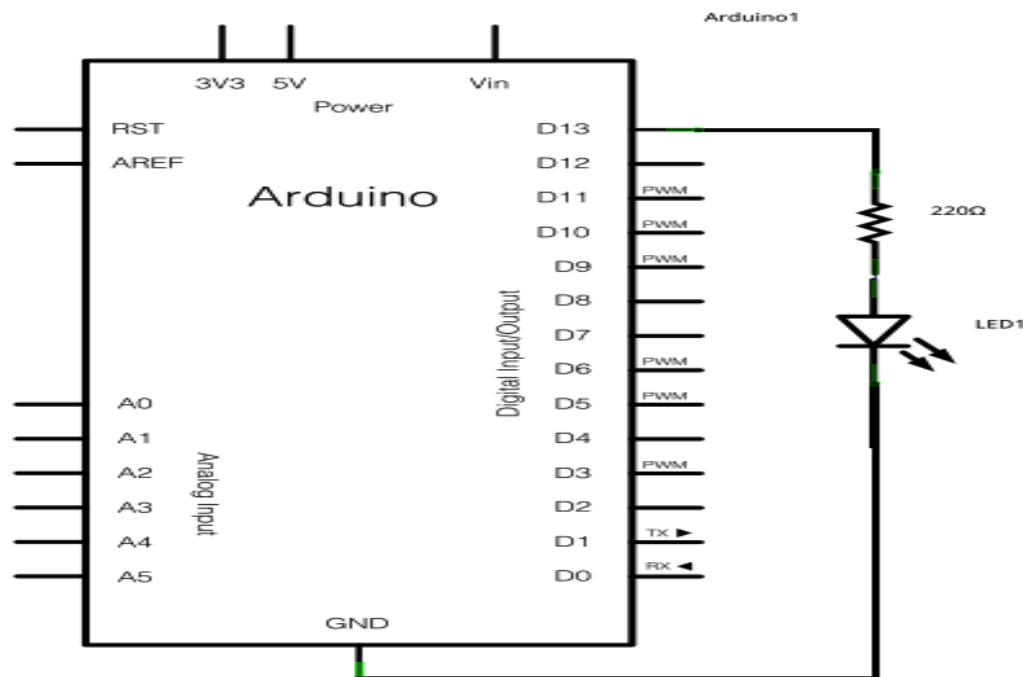
Code:

```
int led = 13;  
void setup() { pinMode(led, OUTPUT); }  
void loop() {  
  digitalWrite(led, HIGH);  
  delay(1000);  
  digitalWrite(led, LOW);  
  delay(1000);  
}
```



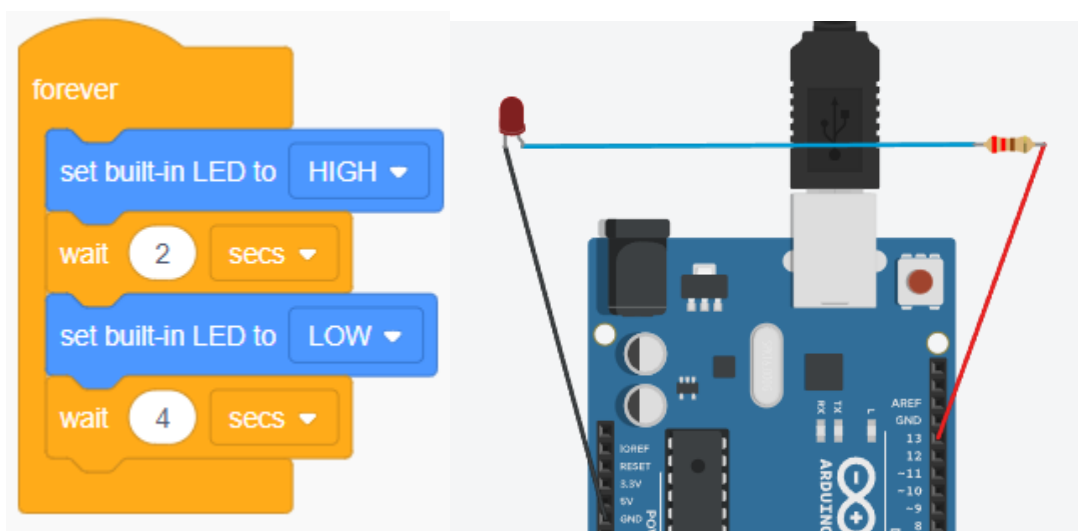
This one shows that it blinks every second

2. Now build a circuit as shown in diagram below:



To build the circuit, attach a 220-ohm resistor to pin 13. Then attach the long leg of an LED (the positive leg, called the anode) to the resistor. Attach the short leg (the negative leg, called the cathode) to ground. Then plug your Arduino board into your computer, start the Arduino program, and enter the code provided previously.

3. Now, try to modify the program so that the LED blinks after a long delay. Explain the syntax that you have modified.



This one makes the LED last for 2 seconds and wait for 4 seconds